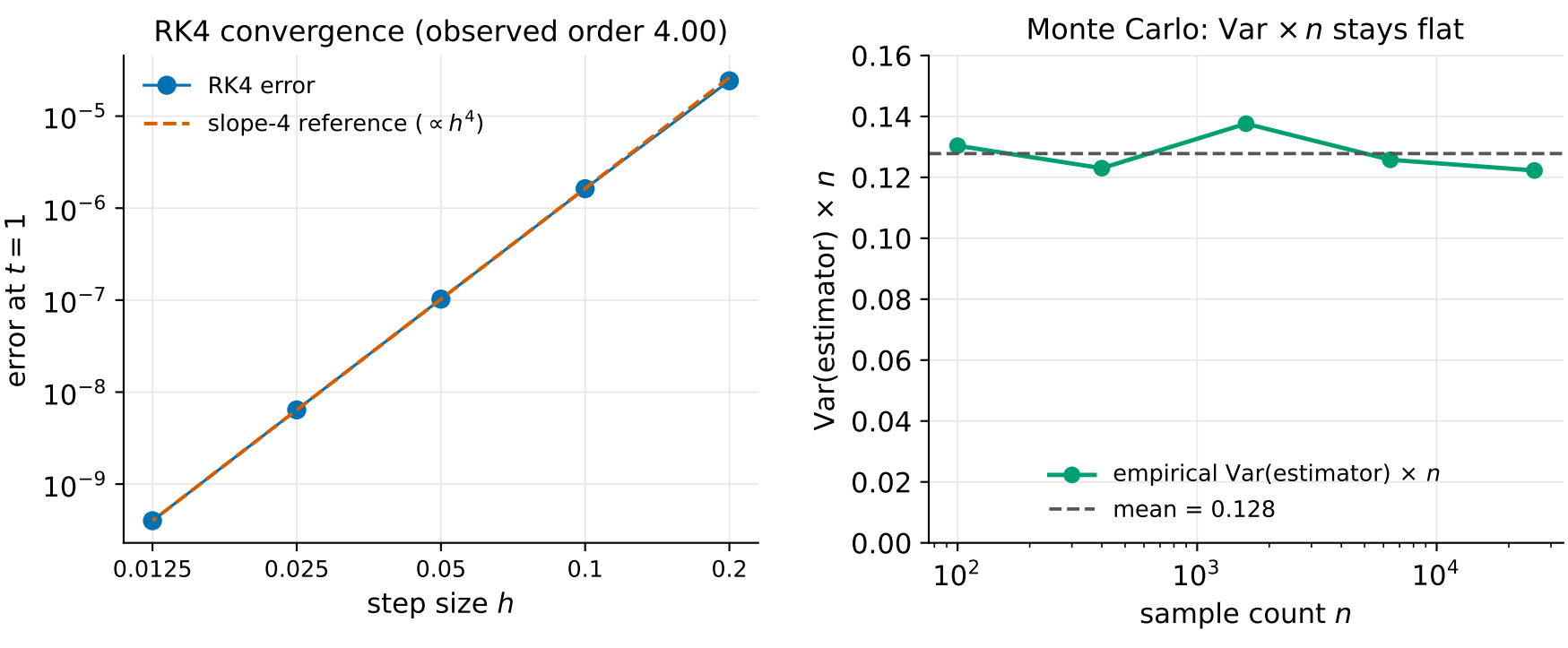
RK4 convergence (observed order 4.00)
RK4 error
slope-4 reference ($\propto h^4$)
error at $t = 1$
10^{-5}
10^{-6}
10^{-7}
10^{-8}
10^{-9}
0.0125
0.025
0.05
0.1
0.2
step size h
Monte Carlo: Var $\times n$ stays flat
Var(estimator) $\times n$
0.16
0.14
0.12
0.10
0.08
0.06
0.04
0.02
0.00
empirical Var(estimator) $\times n$
mean = 0.128
10^2
10^3
10^4
sample count n